ALGEBRA 2 · POLYNOMIAL FUNCTIONS

Zeros & Multiplicity Builder

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Zeros & Multiplicity Builder** on Mr. K's Classroom: Algebra 2 → Polynomial Functions → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. What do you think the graph does at a zero that appears twice, like $(x - 2)^2$?

   It touches the x-axis and turns around (bounces).

2 Explore use the tool and record what you see

Build each polynomial

Function	Degree	End behavior (left, right)	At each zero
$f(x) = (x + 3)(x - 1)(x - 4)$	**3**	**down, up**	**crosses at –3, 1, 4**
$f(x) = (x - 2)^2(x + 1)$	**3**	**down, up**	**bounces at 2, crosses at –1**
$f(x) = -(x + 1)^3$	**3**	**up, down**	**flattens and crosses at –1**
$f(x) = x^2(x - 3)^2$	**4**	**up, up**	**bounces at 0 and 3**
$f(x) = -(x - 1)(x + 2)^2$	**3**	**up, down**	**crosses at 1, bounces at –2**

3 Write the rule fill in each blank from what you observed

1. At a zero with odd multiplicity the graph **crosses** the x-axis. With even multiplicity it **bounces (touches and turns)**.
2. Multiplicity 3 or more makes the graph **flatten** near the zero.
3. The degree is the **sum** of the multiplicities.
4. Even degree: both ends point the **same** way. Odd degree: the ends point **opposite** ways.
5. A negative leading coefficient makes the right end point **down**.

4 Check no tool: show your work

1. Describe $f(x) = (x - 5)^2(x + 2)^3$.

   Degree 5; bounces at 5; flattens and crosses at –2; left end down, right end up

2. Write a polynomial with leading coefficient 1, a zero at –1 of multiplicity 2 and a zero at 4.

   $f(x) = (x + 1)^2(x - 4)$

3. Describe the end behavior of $f(x) = -2x^4 + x$.

   Both ends point down

5 Reflect

Why does an even multiplicity make the graph bounce?

A squared factor is never negative, so the sign of f(x) doesn't change at that zero.